

NETWORK CABLE & CONNECTORS

1) Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

RJ-45 Connector (Front View, Clip Away)

Pin: 1 2 3 4 5 6 7 8

 Wire: W/O O W/G B W/B G W/Br Br

Legend:

W/O = White/Orange
 O = Orange
 W/G = White/Green
 B = Blue
 W/B = White/Blue
 G = Green
 W/Br = White/Brown
 Br = Brown

2) Cable Type	Device/Application	Why Used	
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Cat5e	Wi-Fi Router	Supports up to 1 Gbps speed and is affordable.	
Cat6	Zomato POS Terminal	Faster data transfer and less interference than Cat5e.	
Fiber Optic	IPL Streaming Server	Very high speed and long-distance data transmission.	

3) Steps to Crimp an RJ-45 Connector

- ➔ Cut the network cable to the required length.
- ➔ Remove about 1 inch of the outer cable jacket.
- ➔ Untwist and arrange the wires in the correct color order (T568B is common).
- ➔ Straighten the wires and trim them evenly.
- ➔ Insert the wires into the RJ-45 connector.
- ➔ Check that all wires reach the end of the connector.
- ➔ Place the connector into the crimping tool.
- ➔ Press the crimping tool firmly to secure the connector.
- ➔ Test the cable using a cable tester.

4. Gaming Zone for IPL Screening Event

Cable: Cat6 Ethernet Cable

Connector: RJ-45 Connector

Reason:

- Supports high-speed internet for gaming and IPL streaming.
- Reliable and stable connection with low lag.
- Easy to connect multiple screens, consoles, switches, and routers.

5) Photo of a Network Cable/Connector in Use

Example Description:

Type: Cat6 Ethernet Cable with RJ-45 Connector

Location: Behind a Wi-Fi router

Connected Devices: Wi-Fi router and broadband modem

Purpose: Provides internet connectivity from the modem to the router.

Sample Caption:

This is a Cat6 Ethernet cable with an RJ-45 connector. It connects a broadband modem to a WiFi router and provides a stable internet connection.